

LAB EXPERIMENT RECORD

Set 20: Stock Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze stock price data of three companies using a line chart, bar chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

PROGRAM 1: Line Chart of Stock Prices

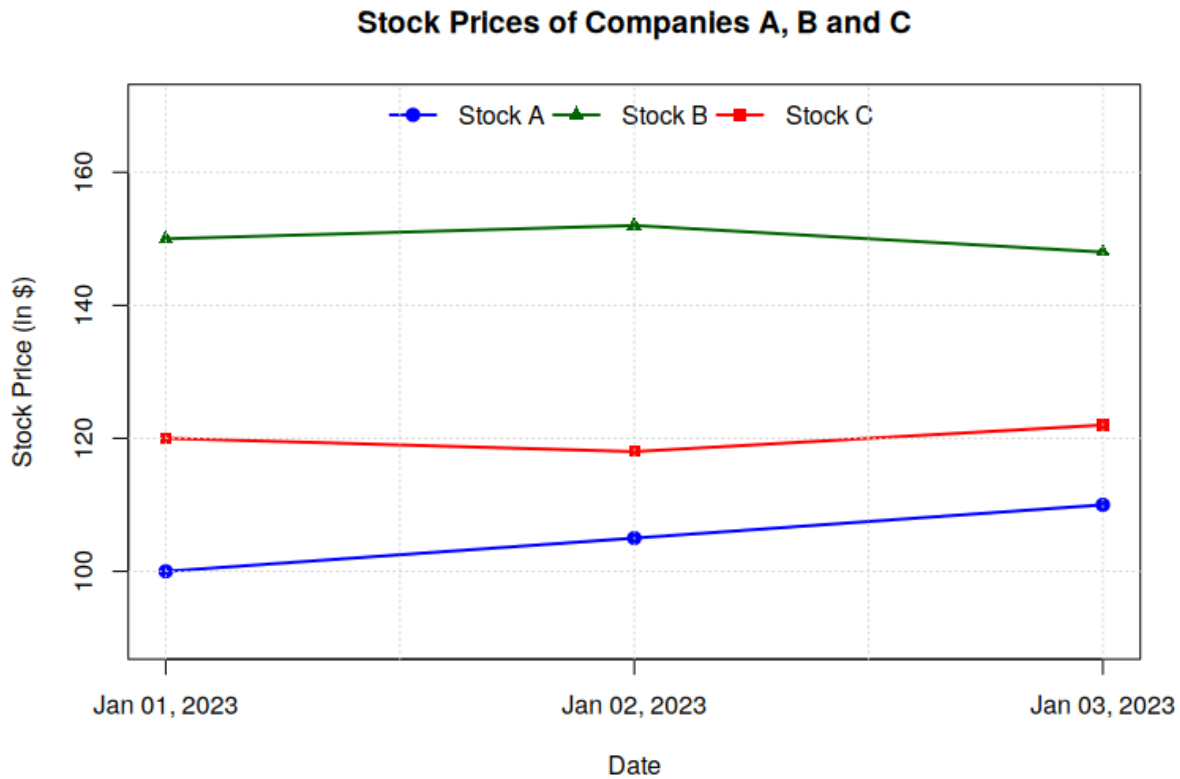
Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

```
# Create dataset for Set 20
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

print(stock_data)

# 1. Line Chart of Stock Prices
stock_cols <- c("blue", "darkgreen", "red")
plot(stock_data$Date, stock_data$Stock_A,
     type = "o", col = stock_cols[1], lwd = 2, pch = 19,
     xaxt = "n", ylim = c(90, 170),
     xlab = "Date", ylab = "Stock Price (in $)",
     main = "Stock Prices of Companies A, B and C")
lines(stock_data$Date, stock_data$Stock_B, type = "o", col = stock_cols[2], lwd = 2, pch = 17)
lines(stock_data$Date, stock_data$Stock_C, type = "o", col = stock_cols[3], lwd = 2, pch = 15)
```

```
axis(1, at = stock_data$Date, labels = format(stock_data$Date, "%b %d, %Y"))
legend("top", legend = c("Stock A", "Stock B", "Stock C"),
      col = stock_cols, pch = c(19, 17, 15), lwd = 2, horiz = TRUE, bty = "n")
grid()
```



Output 1: Stock Prices of Companies A, B and C

Observation: Stock A rises steadily from \$100 to \$110, a gain of 10% over the period. Stock B has the highest price but falls from \$152 to \$148 on the last day. Stock C dips to \$118 on 02 January and then recovers to \$122. Stock A shows the strongest upward trend.

PROGRAM 2: Bar Chart of Daily % Change in Stock A

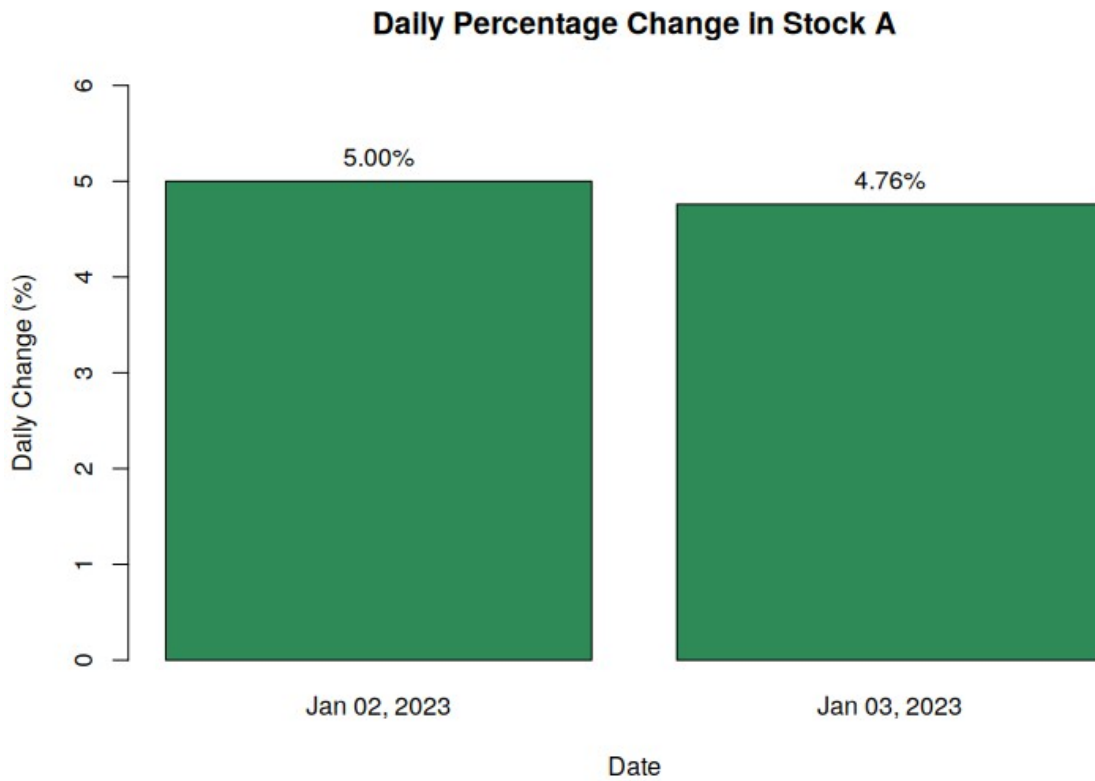
Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

```
# Create dataset for Set 20
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

print(stock_data)

# Daily percentage change for Stock A
stock_data$Change_A <- c(NA, round(100 * diff(stock_data$Stock_A) /
                                           head(stock_data$Stock_A, -1), 2))
print(stock_data[, c("Date", "Stock_A", "Change_A")])

# 2. Bar Chart of Daily % Change in Stock A
change <- stock_data[!is.na(stock_data$Change_A), ]
bp <- barplot(change$Change_A,
              names.arg = format(change$Date, "%b %d, %Y"),
              col = ifelse(change$Change_A >= 0, "seagreen", "red"),
              xlab = "Date",
              ylab = "Daily Change (%)",
              main = "Daily Percentage Change in Stock A",
              ylim = c(0, 6))
text(bp, change$Change_A, labels = paste0(format(change$Change_A, nsmall = 2), "%"), pos
     = 3)
```



Output 2: Daily Percentage Change in Stock A

Observation: Stock A gained 5.00% on 02 January and 4.76% on 03 January, so both days show a positive change. The price increased by the same \$5 on both days, but the percentage is slightly lower on the second day because the starting price was higher.

PROGRAM 3: Table of Stock Price Data

Build a table to display the stock price data for each company over the given period. Label the table elements.

```
# Create dataset for Set 20
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

print(stock_data)

# 3. Table of stock prices
stock_table <- stock_data
stock_table$Date <- format(stock_data$Date, "%Y-%m-%d")
colnames(stock_table) <- c("Date", "Stock A ($)", "Stock B ($)", "Stock C ($)")

cat("=== STOCK PRICE TABLE ===\n")
print(stock_table, row.names = FALSE)
```

```
=== STOCK PRICE TABLE ===
> print(stock_table, row.names = FALSE)
      Date Stock A ($) Stock B ($) Stock C ($)
2023-01-01      100      150      120
2023-01-02      105      152      118
2023-01-03      110      148      122
>
```

Output 3: Stock Price Table

Observation: The table shows the daily prices of all three stocks with clearly labelled columns. Stock B is the highest-priced stock on every day, while Stock A is the lowest-priced but the fastest-growing.

PROGRAM 4: Interactive Dashboard for Stock Prices

Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 20
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)
stock_cols <- c(Stock_A = "blue", Stock_B = "darkgreen", Stock_C = "red")

ui <- fluidPage(
  titlePanel("Stock Price Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("stocks", "Stocks on line chart:",
        choices = c("Stock A" = "Stock_A", "Stock B" = "Stock_B", "Stock
C" = "Stock_C"),
        selected = names(stock_cols)),
      selectInput("change_stock", "Daily % change for:",
        choices = c("Stock A" = "Stock_A", "Stock B" = "Stock_B", "Stock C" =
"Stock_C"),
        selected = "Stock_A")
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "300px")),
        column(6, plotOutput("barPlot", height = "300px"))
      ),
      h4("Stock Price Table"),
      tableOutput("stockTable")
    )
  )
)

server <- function(input, output) {
  output$linePlot <- renderPlot({
    plot(NA, xlim = range(stock_data$Date), ylim = c(90, 170), xaxt = "n",
      xlab = "Date", ylab = "Stock Price ($)", main = "Stock Prices Over Time")
    axis(1, at = stock_data$Date, labels = format(stock_data$Date, "%b %d"))
    for (s in input$stocks) {
      lines(stock_data$Date, stock_data[[s]], type = "o", col = stock_cols[s], lwd = 2,
pch = 19)
    }
    if (length(input$stocks) > 0)
      legend("top", legend = gsub("_", " ", input$stocks), col =
stock_cols[input$stocks],
        lwd = 2, horiz = TRUE, bty = "n", cex = 0.8)
  })
}
```

```

    grid()
  }, width = 420, height = 300, res = 96)

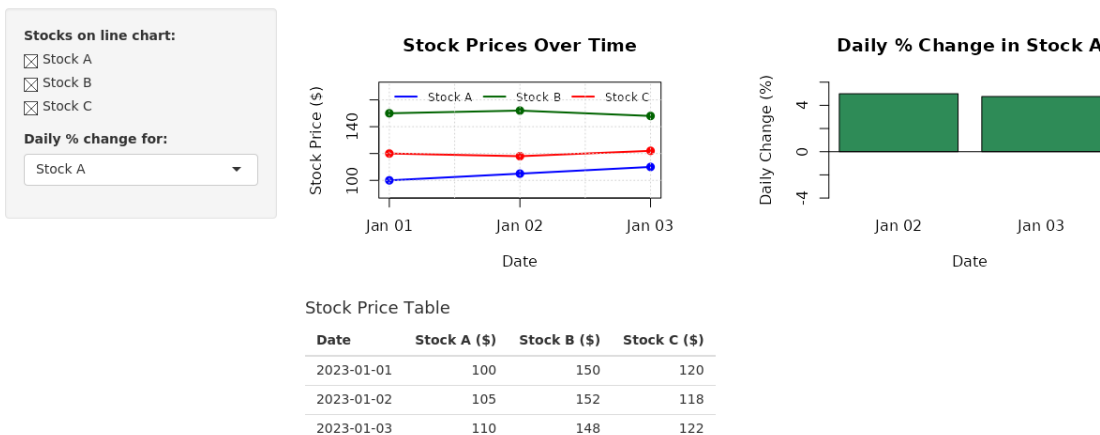
output$barPlot <- renderPlot({
  p <- stock_data[[input$change_stock]]
  change <- round(100 * diff(p) / head(p, -1), 2)
  barplot(change, names.arg = format(stock_data$Date[-1], "%b %d"),
    col = ifelse(change >= 0, "seagreen", "red"), ylim = c(-4, 6),
    xlab = "Date", ylab = "Daily Change (%)",
    main = paste("Daily % Change in", gsub("_", " ", input$change_stock)))
  abline(h = 0)
}, width = 420, height = 300, res = 96)

output$stockTable <- renderTable({
  d <- stock_data
  d$Date <- format(d$Date, "%Y-%m-%d")
  colnames(d) <- c("Date", "Stock A ($)", "Stock B ($)", "Stock C ($)")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Stock Price Dashboard



Output 4: Stock Price Interactive Dashboard (R Shiny)

Observation: The dashboard shows the stock price line chart, the daily percentage change bar chart, and the stock price table together. The checkboxes choose which stocks appear on the line chart, and the drop-down list selects the stock whose daily change is shown. In the screenshot, Stock A gains 5.00% and 4.76% on the two days.

Procedure

1. Create the stock price data frame with Date and the prices of Stock A, Stock B, and Stock C.
2. Use `plot()` and `lines()` to draw the line chart for the three stocks, with `axis()` for date labels and `legend()` for the stock names.
3. Calculate the daily percentage change of Stock A using `diff()` and draw the bar chart using `barplot()` with `text()` labels.
4. Format the dates, rename the columns with `colnames()`, and print the labelled stock price table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The stock price data was successfully visualized using a line chart, bar chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>as.Date()</code>	Converts text values into dates.
<code>plot()</code> / <code>lines()</code>	Draws the line chart for the three stocks.
<code>diff()</code>	Calculates the day-to-day price change.
<code>barplot()</code>	Creates the percentage change bar chart.
<code>legend()</code> / <code>axis()</code>	Adds the legend and date labels.
<code>colnames()</code> / <code>print()</code>	Labels and displays the stock table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.